Shutdown Inning Percentage (SIP)

A new pitching statistic for the half-inning after your team scores, tested on 61,634 MLB games, 2000-2025

Abstract. Baseball people routinely praise the "shutdown inning," the scoreless half-inning a pitcher throws right after his team scores. We formalize this as **Shutdown Inning Percentage (SIP)**: scoreless half-innings divided by shutdown opportunities, where an opportunity is any half-inning that immediately follows a half-inning in which the pitcher's team scored. Using Retrosheet play-by-play for every regular-season game from 2000 to 2025 (1.10 million half-innings; 291,338 shutdown opportunities) we find that the league-wide SIP is 72.1%, statistically indistinguishable from the 72.2% scoreless rate in all other half-innings (adjusted odds ratio of allowing a run 1.003, 95% CI 0.988-1.019). The number of runs the team just scored does not matter. SIP carries some real information about pitchers: elite pitchers such as Mariano Rivera, Clayton Kershaw and Jacob deGrom top the leaderboard. But that information is general run prevention. The portion of SIP that is separate from a pitcher's ordinary innings has a split-half reliability near zero (r = -0.04), and we find only a small "coasting" effect with leads of five or more runs (about 0.8 points within the same pitchers). SIP is a clean descriptive statistic that belongs next to other context stats in a box score; the evidence does not support it as a separate measure of mental toughness.

1. Introduction

Managers and broadcasters talk about the shutdown inning as if it were a skill: after an offense puts up a crooked number, the pitcher is expected to "go out there and shut the door." Traditional statistics do not measure this. Wins credit a pitcher merely for being the pitcher of record when his team scores; ERA and WHIP are indifferent to when runs are allowed. A pitcher who gives back a four-run lead immediately and a pitcher who protects it can have identical lines.

This paper (i) defines SIP precisely, (ii) builds a public dataset of every shutdown opportunity from 2000 to 2025, and (iii) tests three claims that are implicit in the baseball folklore: *the shutdown inning is a distinct situation* (it is harder or easier than other innings), *bigger run support makes pitchers coast*, and *some pitchers are consistently better at it than their ordinary performance predicts*.



Figure 1. Definition of a shutdown opportunity and of SIP.

2. Definition and data

Shutdown opportunity. A half-inning is a shutdown opportunity for a team's pitcher if the half-inning played immediately before it was a half-inning in which that team batted and scored one or more runs. The pitcher who throws the first pitch of the half-inning is credited with the opportunity. A **shutdown** is an opportunity in which the opponent scores zero runs. **SIP = shutdowns / opportunities x 100.**

Data. We use Retrosheet regular-season event files (as maintained by the Chadwick Baseball Bureau) for 2000-2025, parsed with the Chadwick *cwevent* tool. Runs on each play are counted from the batter and runner destinations. Half-innings are rebuilt from 4.86 million events. The game total of 8.9-9.7 combined runs per game in each season matches published scoring levels. Totals: 61,634 games, 1,101,190 half-innings, 291,338 shutdown opportunities (26.5% of half-innings) and 748,218 comparison half-innings (all non-first half-innings that are not opportunities). The first half-inning of a game cannot be an opportunity and is excluded from comparisons.

Conventions. (1) Credit goes to the pitcher who starts the half-inning, even if a reliever finishes it; the success test uses all runs in the half-inning. (2) Walk-off innings count: a home team that wins in the bottom half is a failure for the visiting pitcher. (3) Extra innings from 2020 onward start with an automatic runner on second; they are flagged and modeled. (4) Playoffs and the All-Star Game are excluded.

3. Methods

Context model. To judge how hard an opportunity was, we fit a logistic regression for scoreless half-innings using only pitcher-independent factors: the opposing offense in that season, the ballpark in a five-year era, inning, home/away and the extra-inning runner rule. Its prediction, the expected scoreless probability, gives each pitcher an *expected SIP*. **SIP+** is SIP minus expected SIP.

Adjusted effects. We estimate odds ratios for allowing a run with logistic regression adjusted for season, inning, home/away, starter vs reliever, pitching-team and batting-team effects, with standard errors clustered by game. Model 1 compares opportunities to other half-innings (400,000 randomly sampled half-innings). Model 2 looks only inside opportunities and compares 2, 3, 4 and 5+ runs scored against 1 run. Model 3 groups the lead entering the inning.

Pitcher skill. We split each pitcher's innings into odd and even halves in chronological order, correlate the halves, and apply the Spearman-Brown correction. We also compute the season-to-season correlation, the spread of career SIP compared with a binomial luck benchmark, and an empirical-Bayes estimate of true talent variance by method of moments. The key quantity is the **gap** between a pitcher's SIP and his scoreless rate in all other half-innings: if "clutch" shutdown ability exists, the gap should repeat.

Win probability. A gradient-boosted model of win probability by lead, inning and batting side (from all 61,634 games) values each opportunity by the change in the pitching team's win probability across the half-inning.

4. Results

4.1 League level: no shutdown effect

The league SIP is **72.08%**; all other half-innings are scoreless **72.17%** of the time, and the context model expects 72.00%. The two series move together over 26 seasons (Figure 2) and the yearly gaps are small and

of both signs, even as the scoreless rate ranged from 68.7% to 74.4% across eras. After adjustment, a shutdown opportunity changes the odds of allowing a run by a factor of 1.003 (95% CI 0.988-1.019, p = 0.66). Figure 4 shows the same pattern inning by inning, with extra innings the one visible exception.

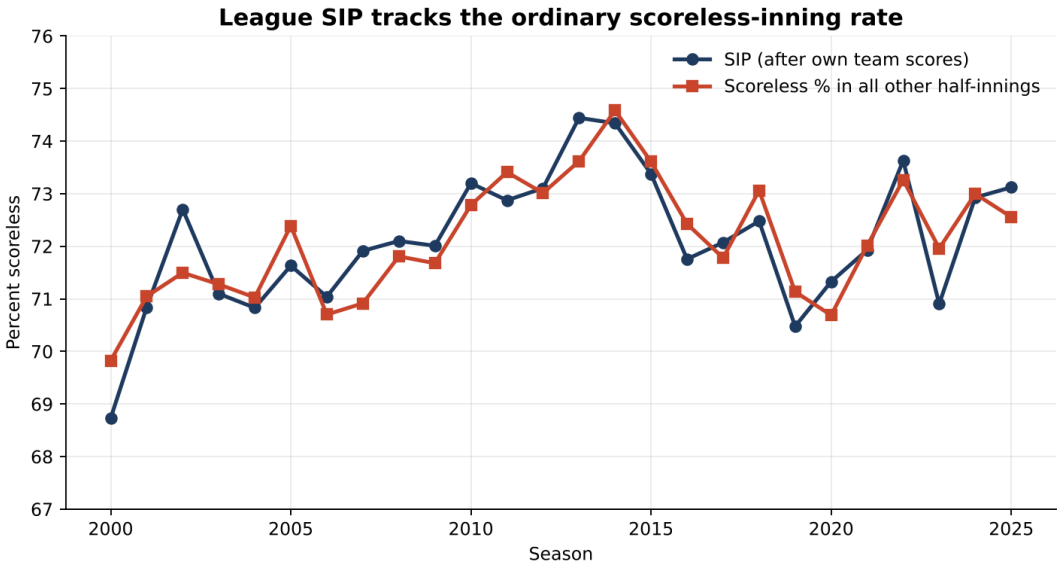


Figure 2. League SIP and the scoreless rate in all other half-innings, 2000-2025.

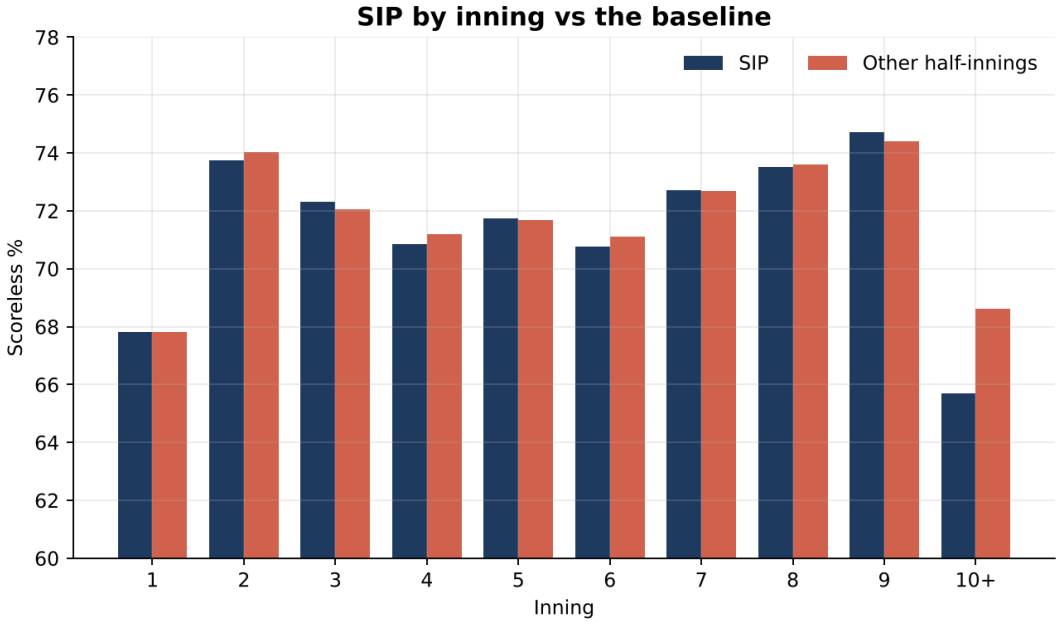


Figure 4. SIP and the baseline by inning. Both series share the same inning-by-inning shape.

4.2 The size of the cushion does not matter

If momentum made big innings harder to protect, SIP would fall as the prior half-inning gets bigger. It does not. SIP is 72.1% after one run, 72.2% after two, 71.9% after three, 72.1% after four and 71.9% after five or more (Figure 3). Adjusted odds ratios against a one-run inning are 0.995, 1.017, 1.018 and 1.032, all with confidence intervals that include 1 (Figure 5). A long wait on the bench, proxied by eight or more batters in the preceding half-inning, has no clear effect (OR 0.977, 0.942-1.013).

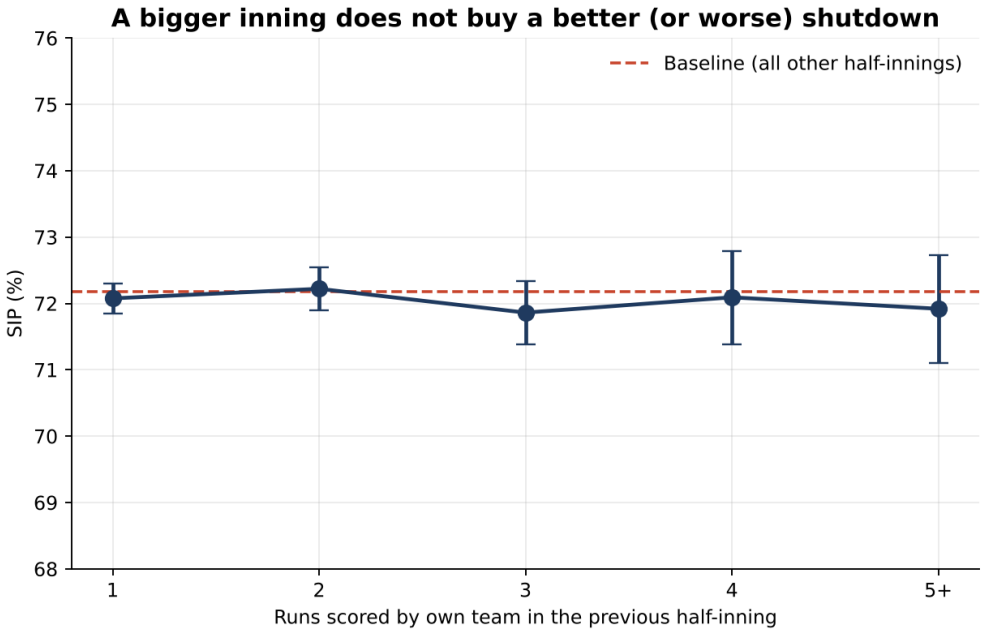


Figure 3. SIP by runs scored in the preceding half-inning, with 95% Wilson intervals.

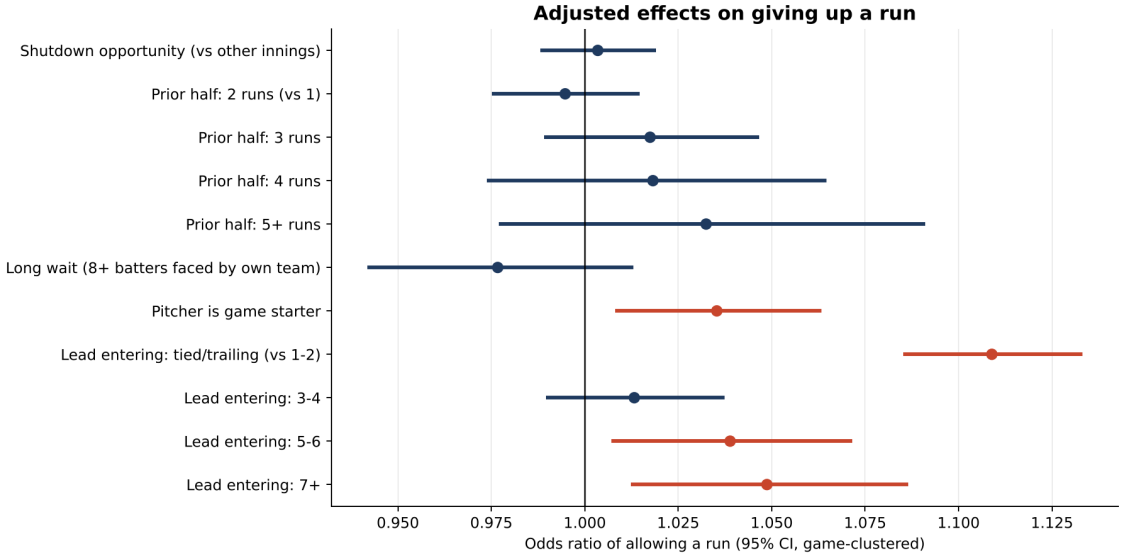


Figure 5. Adjusted odds ratios of allowing a run (game-clustered 95% intervals). Red intervals exclude 1.

4.3 A small coasting effect with big leads

Lead size is a different question from the size of the last inning: a pitcher can be up 7-1 after a one-run inning. Here a small effect appears. Entering the inning with a lead of 5-6 runs raises the odds of allowing a run by 3.9% (OR 1.039, 1.007-1.072) and a lead of 7 or more by 4.9% (OR 1.049, 1.012-1.087) relative to a 1-2 run lead. Because managers hand blowout innings to worse relievers, the raw effect mixes coasting with selection. Comparing the same pitchers across situations removes selection: among 530 pitchers with at least 25 opportunities in each group, SIP with a lead of 5+ runs is **0.76 points lower** than with a 1-2 run lead after context adjustment (SE 0.32). This is real but small, about one scoreless inning in 130. Pitchers who are tied or trailing despite just scoring do worse (OR 1.109), which more likely reflects weaker pitching than momentum.

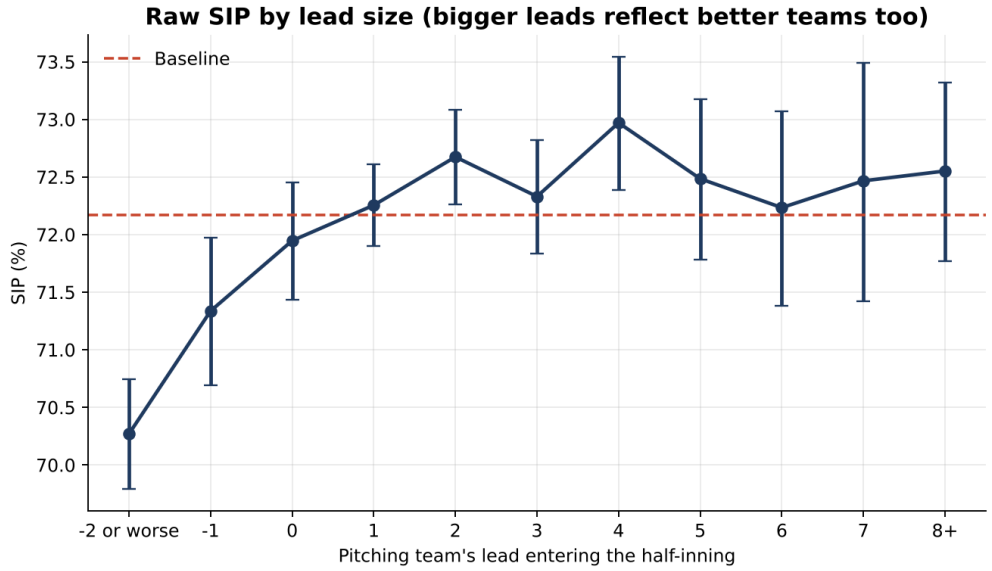


Figure 6. Raw SIP by the pitching team's lead entering the half-inning.

4.4 Is SIP a pitcher skill?

Pitcher SIP spreads out only modestly more than chance would produce. Among 887 pitchers with 100 or more opportunities, the observed standard deviation of SIP is 4.02 points against 3.33 expected from binomial luck (Figure 7). The implied true-talent standard deviation of SIP+ is only 1.9 points (1,457 pitchers with 50+ opportunities). Split-half reliability (Figure 8, 1,297 pitchers) is 0.15 for SIP (0.26 after Spearman-Brown), compared with 0.41 (0.58) for the pitcher's scoreless rate in his other half-innings. The context-adjusted SIP+ has reliability 0.08 and the gap between SIP and the pitcher's own baseline has reliability **-0.04**. Year to year, SIP correlates at 0.12, the ordinary scoreless rate at 0.28, and the gap at -0.00 (2,981 pitcher-season pairs).

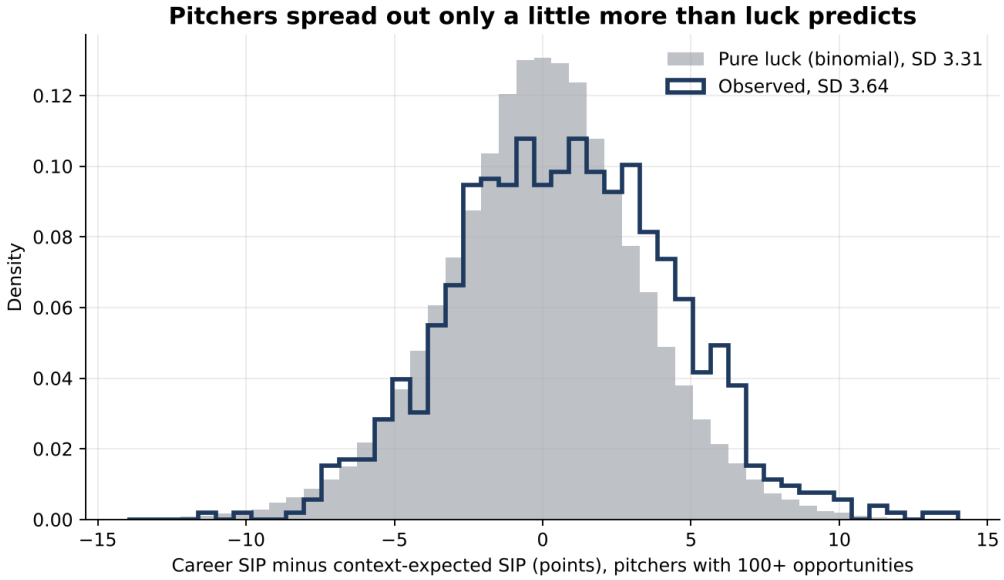


Figure 7. Spread of career SIP minus expected SIP against a binomial luck benchmark (pitchers with 100+ opportunities).

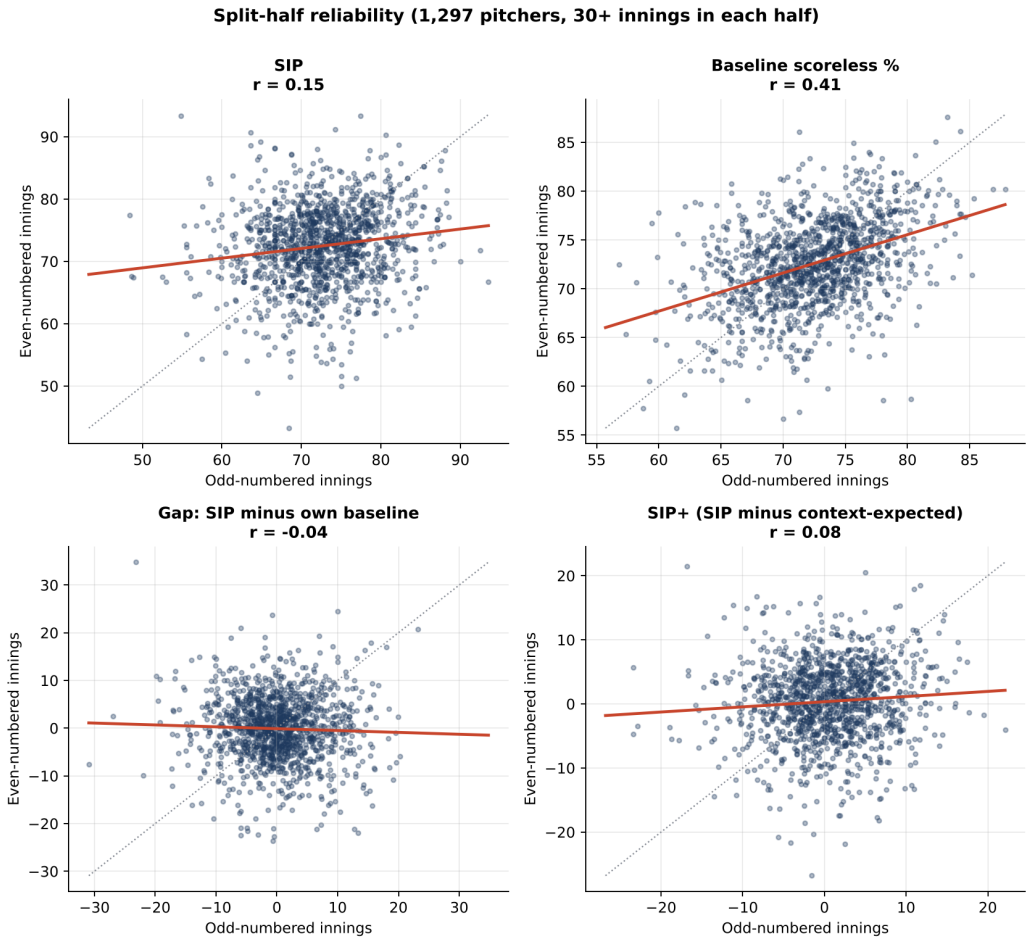


Figure 8. Split-half reliability of SIP, the baseline, the gap and SIP+.

The pitcher-level correlation between SIP and the scoreless rate in other innings is **0.42** (Figure 9), and at the team-season level it is 0.56 (Figure 12). In other words, the pitchers who are good at shutdown innings are the pitchers who are good at all innings; the shutdown-specific remainder does not repeat.

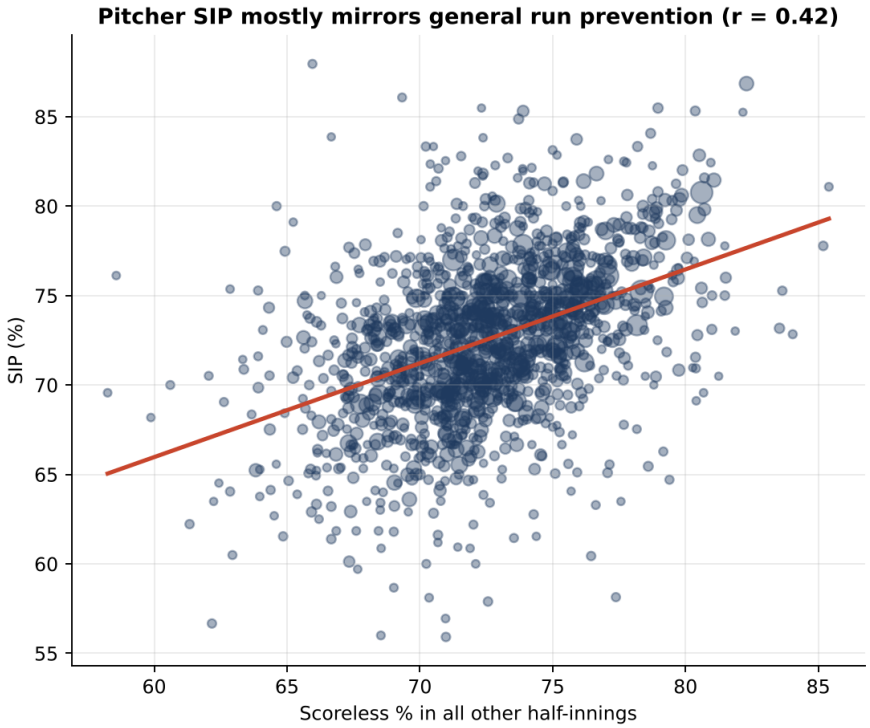


Figure 9. Pitcher SIP vs scoreless % in other half-innings (60+ innings in each group; point size is opportunities).

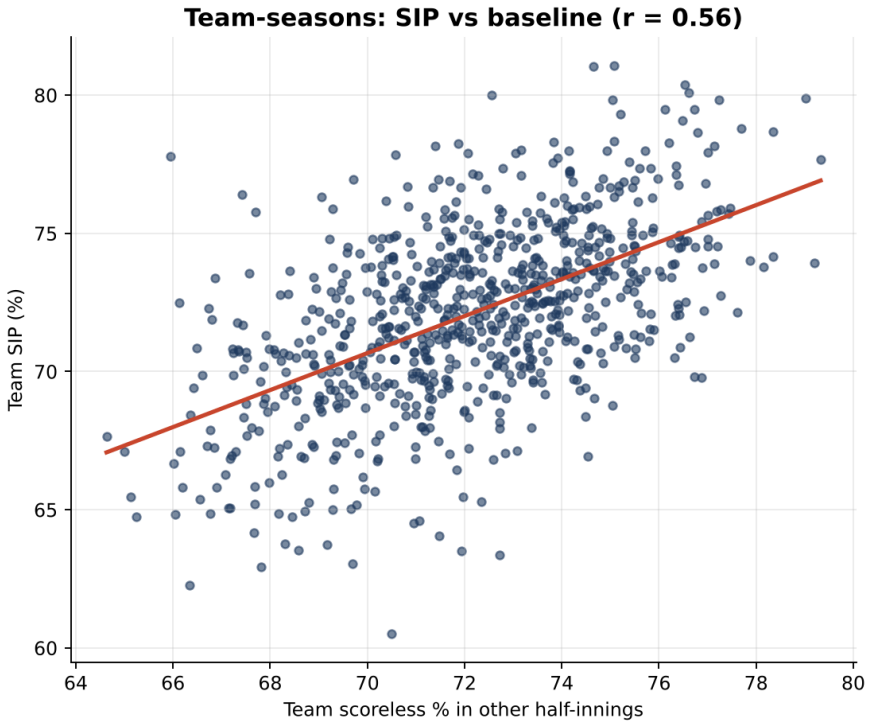


Figure 12. Team-season SIP vs baseline.

4.5 Leaders and laggards

Table 1 lists the career extremes with 200+ opportunities after regression toward expectation. The leaders are the best pitchers of the era, mostly closers and aces. The bottom list is mostly back-of-the-rotation

starters. Regression matters: Rivera's raw 86.9% is pulled to 79.8% because 289 opportunities is not enough to treat the whole gap above expectation as skill.

Pitcher	Years	Opps	SIP	Expected	Regressed
Mariano Rivera	2000-2013	289	86.9	73.1	79.8
Clayton Kershaw	2008-2025	759	80.8	73.5	78.2
Francisco Rodriguez	2002-2017	297	81.8	74.1	77.4
Billy Wagner	2000-2010	204	82.8	73.7	76.9
Tim Lincecum	2007-2016	423	80.1	73.5	76.8
Kenley Jansen	2010-2025	270	80.4	74.3	76.6
Jacob deGrom	2014-2025	400	79.5	73.5	76.4
Joe Nathan	2000-2016	275	81.5	72.7	76.3

Table 1a. Highest regressed career SIP (200+ opportunities).

Pitcher	Years	Opps	SIP	Expected	Regressed
Rob Bell	2000-2007	205	62.9	69.0	67.5
Shawn Chacon	2001-2008	257	65.0	68.8	67.7
Antonio Senzatela	2017-2025	210	64.8	68.7	67.7
Jason Johnson	2000-2008	294	63.6	70.2	68.1
Esteban Loaiza	2000-2008	412	65.5	70.0	68.2
Josh Fogg	2001-2009	330	64.5	70.4	68.4
Steve Sparks	2000-2004	200	65.0	69.7	68.6
Mike Maroth	2002-2007	274	66.1	69.9	68.7

Table 1b. Lowest regressed career SIP.

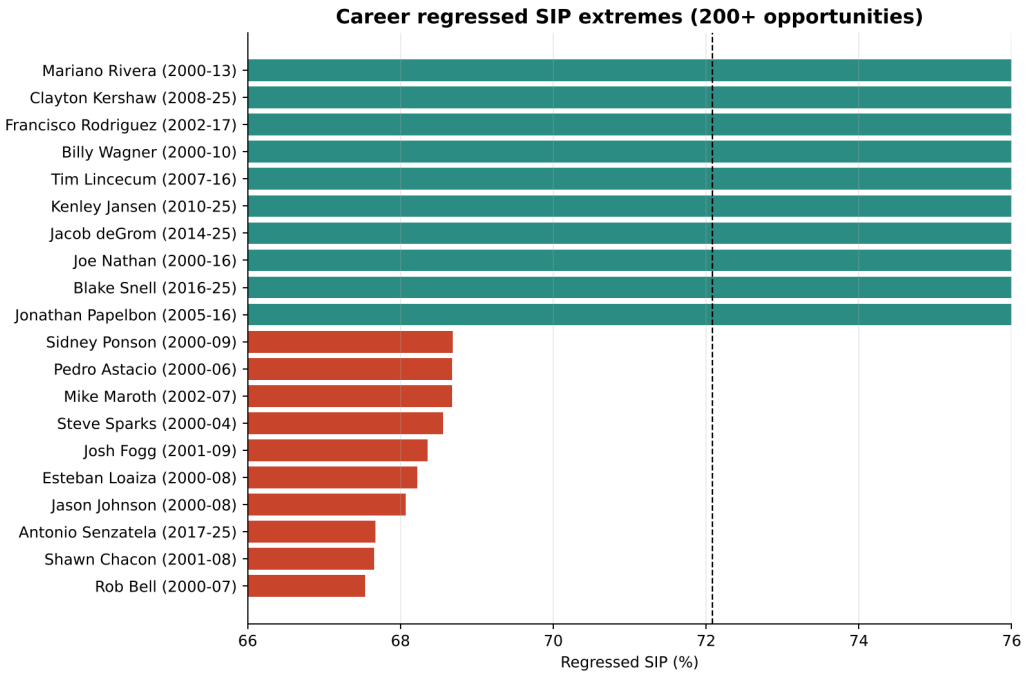


Figure 11. Career regressed SIP extremes.

4.6 What a shutdown is worth

Whatever its cause, the shutdown inning matters for winning. After an opportunity, a scoreless inning raises the pitching team's win probability by about 5.0 points on average; allowing a run lowers it by about 13.1 points, an 18-point swing (Figure 10). The swing is similar across innings 1-9 and larger in extra innings.

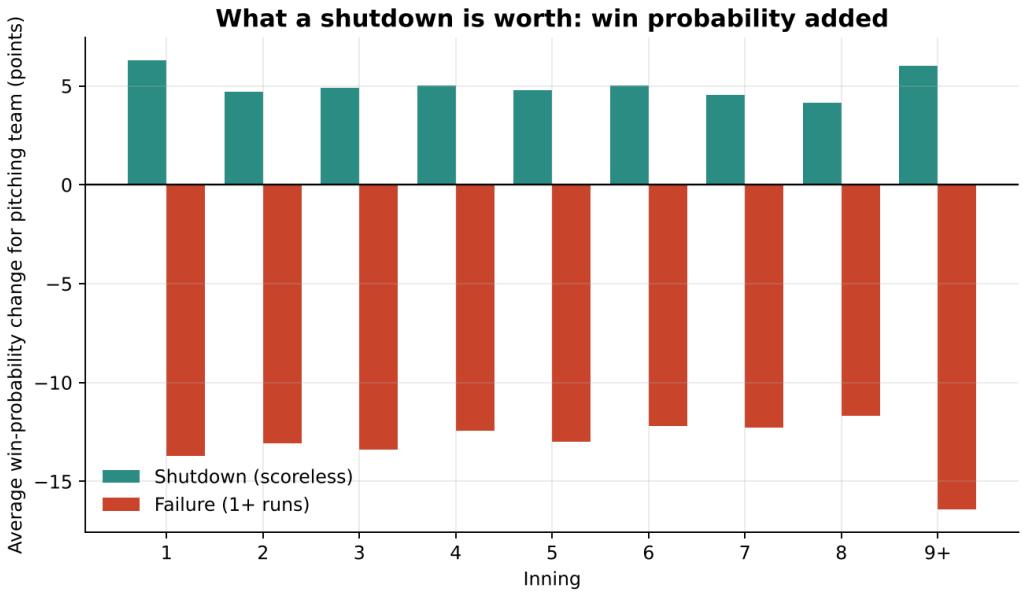


Figure 10. Average win-probability change for the pitching team across an opportunity half-inning, by outcome.

5. Discussion

What SIP is. SIP is a transparent, easy-to-audit descriptive statistic. It tells you how often a pitcher actually protected a lead he was just handed, and it ties to a real swing in win probability. For fans, broadcasters and front offices that want to talk about shutdown innings, SIP gives the phrase a number. It can appear in a box score as a simple line (for example "SD 3/4") at essentially no cost, because every needed input already exists in a play-by-play record.

What SIP is not. The folklore that shutdown innings are a distinct psychological test does not survive the data. The league scores the same share of innings whether or not the home team has just scored, the amount scored does not matter, and the pitcher-specific part of SIP does not repeat. Its predictive value is general talent: pitchers with high SIP are the pitchers with high scoreless rates everywhere. We therefore recommend reporting SIP together with a context benchmark (expected SIP) and treating small samples with caution: at 40 opportunities a pitcher's SIP has a standard error of about 7 points, larger than the true spread among pitchers (about 2 points).

Coasting. Large leads give a small, measurable decline (about 0.8 points within pitchers). That is consistent with pitchers throwing more strikes and trading some run prevention for efficiency when the game is not close, but the data cannot separate that from game-state selection and defensive alignments.

6. Limitations

Credit is assigned to the pitcher who starts the half-inning, so relievers who inherit trouble are not charged for a predecessor's runners and a starter pulled mid-inning may be blamed or credited for what happens after him. SIP+ adjusts for opposing offense, park era, inning and the extra-inning rule but not for the pitcher's own defense, pitch quality or handedness matchups. The empirical-Bayes variance estimate assumes a roughly normal distribution of talent. The win-probability model is empirical and ignores base-out states inside innings. The reliability comparison between SIP and the baseline is not exactly like-for-like because the baseline pools about 2.6 times as many half-innings. The sample is regular-season, 2000-2025; 2020 is a 60-game season.

7. Reproducibility and data

All inputs are public (Retrosheet). The accompanying package contains the shutdown-opportunity dataset, the full half-inning table, pitcher-career and pitcher-season SIP tables, team-season and league tables, regression output, the interactive dashboard and every figure as a separate PNG. The data dictionary lists each column. The information used here was obtained free of charge from and is copyrighted by Retrosheet (www.retrosheet.org).

Appendix: key numbers

Quantity	Value
Games / half-innings / opportunities	61,634 / 1,101,190 / 291,338
League SIP / baseline / context-expected	72.08% / 72.17% / 72.00%
Adjusted OR, opportunity vs other (95% CI)	1.003 (0.988-1.019)
Prior runs 2 / 3 / 4 / 5+ vs 1 (OR)	0.995 / 1.017 / 1.018 / 1.032
Lead 5-6 / 7+ vs 1-2 (OR)	1.039 / 1.049
Within-pitcher big-lead effect, SIP points	-0.76 (SE 0.32, n = 530)
Split-half r: SIP / baseline / gap / SIP+	0.15 / 0.41 / -0.04 / 0.08
Year-over-year r: SIP / baseline / gap	0.12 / 0.28 / -0.00
Win probability: shutdown / failure	+5.0 / -13.1 points